

SHA-512 Hash Function

Practical / Laboratory Program

1. Aim

To implement the SHA-512 (Secure Hash Algorithm 512-bit) hash function and generate a 512-bit message digest for a given input.

2. Algorithm

1. Start the program.
2. Read the input message from the user.
3. Initialize the SHA-512 hash object.
4. Encode the input message into bytes.
5. Update the hash object with the input bytes.
6. Compute the SHA-512 digest.
7. Display the hexadecimal hash value.
8. Stop the program.

3. Program

```
import hashlib

message = input("Enter the message: ")

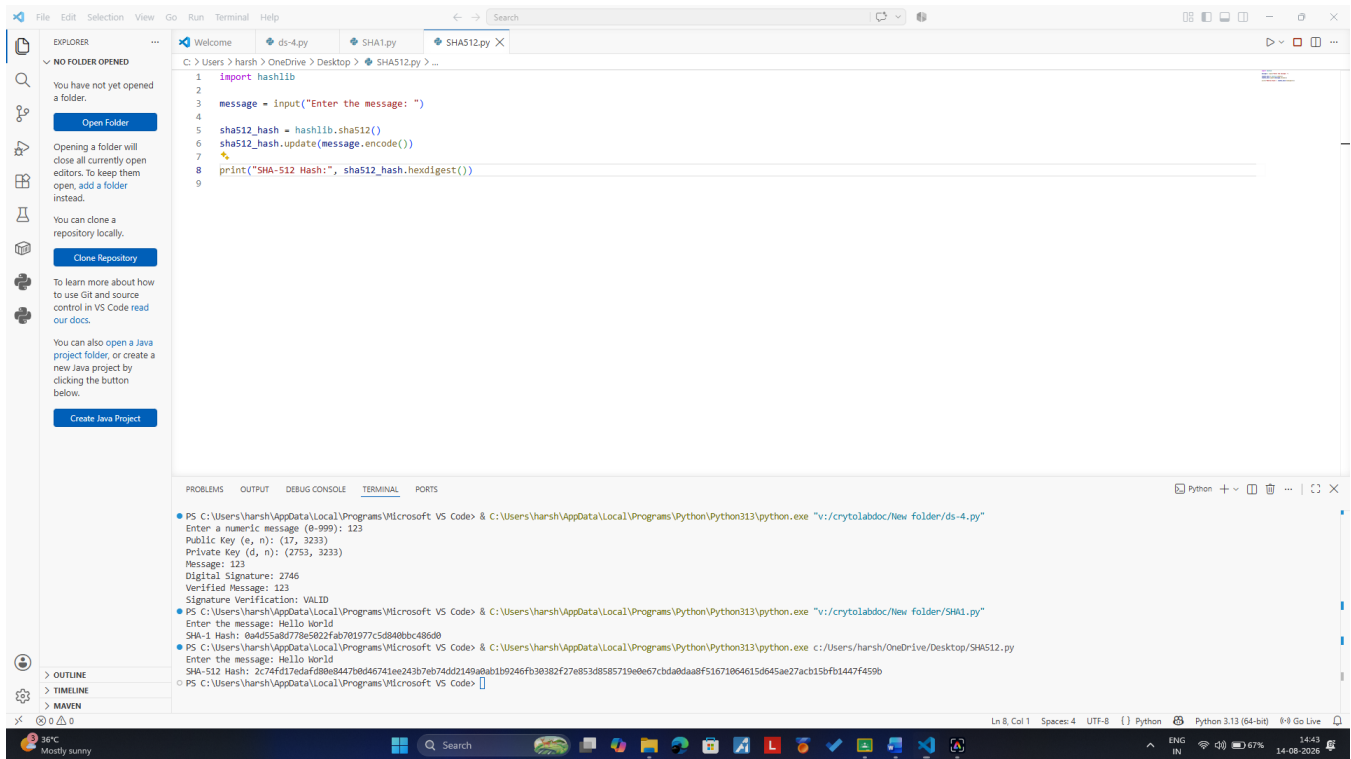
sha512_hash = hashlib.sha512()
sha512_hash.update(message.encode())

print("SHA-512 Hash:", sha512_hash.hexdigest())
```

4. Input

Hello World

5. Output



The screenshot displays the Visual Studio Code interface. The Explorer panel on the left shows a project structure with files 'Welcome', 'ds-4.py', 'SHA1.py', and 'SHA512.py'. The main editor window is open to 'SHA512.py', which contains the following Python code:

```
1 import hashlib
2
3 message = input("Enter the message: ")
4
5 sha512_hash = hashlib.sha512()
6 sha512_hash.update(message.encode())
7
8 print("SHA-512 Hash:", sha512_hash.hexdigest())
9
```

The Output panel at the bottom shows the execution results of the script:

```
PS C:\Users\harsh\AppData\Local\Programs\Microsoft VS Code> & C:\Users\harsh\AppData\Local\Programs\Python\Python313\python.exe "v:/cryptolabdoc/new_folder/ds-4.py"
Enter a numeric message (0-999): 123
Public Key (e, n): (17, 3233)
Private Key (d, n): (2753, 3233)
Message: 123
Digital Signature: 2746
Verified Message: 123
Signature Verification: VALID
PS C:\Users\harsh\AppData\Local\Programs\Microsoft VS Code> & C:\Users\harsh\AppData\Local\Programs\Python\Python313\python.exe "v:/cryptolabdoc/new_folder/SHA1.py"
Enter the message: Hello world
SHA-1 Hash: 8a4d55a8d778e5022fab701977c5d9408bc406d0
PS C:\Users\harsh\AppData\Local\Programs\Microsoft VS Code> & C:\Users\harsh\AppData\Local\Programs\Python\Python313\python.exe c:/Users/harsh/OneDrive/Desktop/SHA512.py
Enter the message: Hello world
SHA-512 Hash: 2c74fd17edaf08be0447bed66741ee243b7eb74dd2149a0ab1b9246fb30382f27e853d8585719e0e67cbda0daa8f51671064615d645ae27ac315bf1447f459b
PS C:\Users\harsh\AppData\Local\Programs\Microsoft VS Code>
```

6. Result

Thus, the SHA-512 hash function was implemented successfully. The program generates a 512-bit (128 hexadecimal character) digest.